ABTechSolutions Enterprise Network Infrastructure

Multi-Site Network with Security & High Availability

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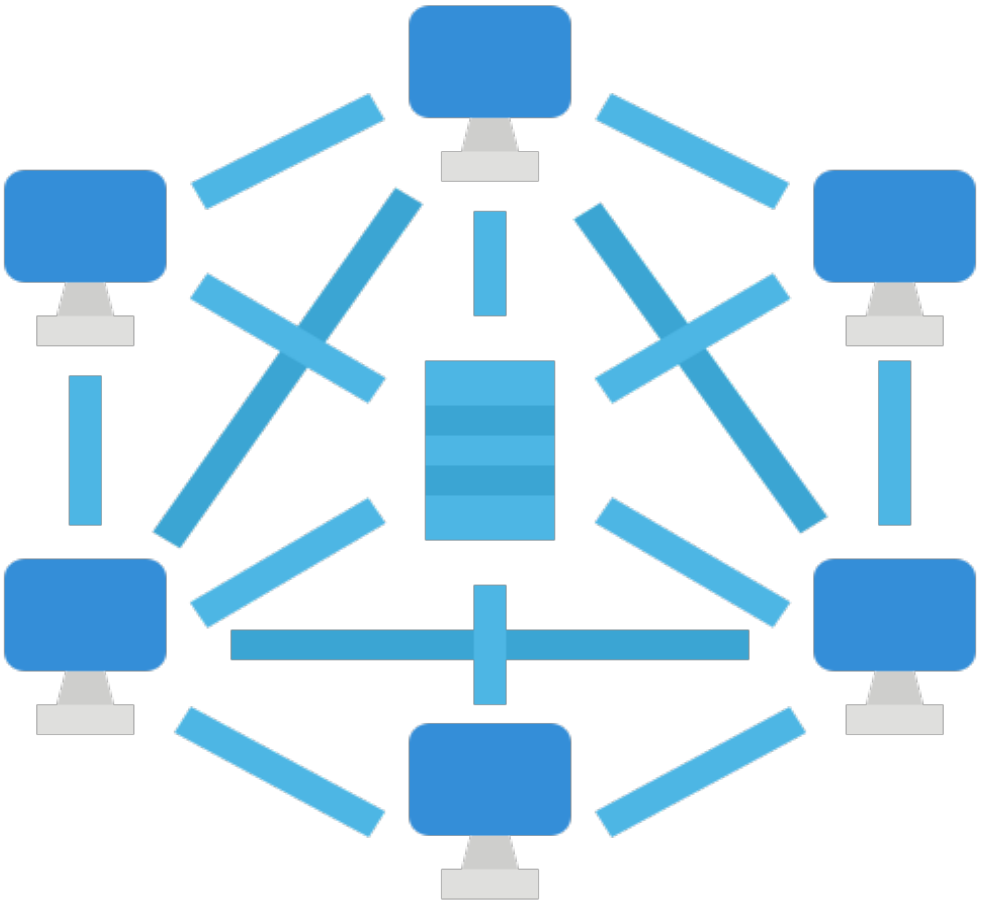
Computer Networks



A comprehensive enterprise network design demonstrating real-world networking concepts including OSPF dynamic routing, HSRP redundancy, VLAN segmentation, DMZ architecture, dual wireless networks, and centralized monitoring across headquarters and two branch offices.

Section 2: PROJECT OVERVIEW

This project demonstrates a complete enterprise network with:

- 3 Sites: HQ + 2 Branch Offices

- 50+ Devices: 6 routers, 6 switches, 9 servers

- 8 VLANs with inter-VLAN routing

- OSPF dynamic routing across all sites

- HSRP redundancy (99.9% uptime)

- Dual wireless networks (internal + guest)

- DMZ with NAT for public services

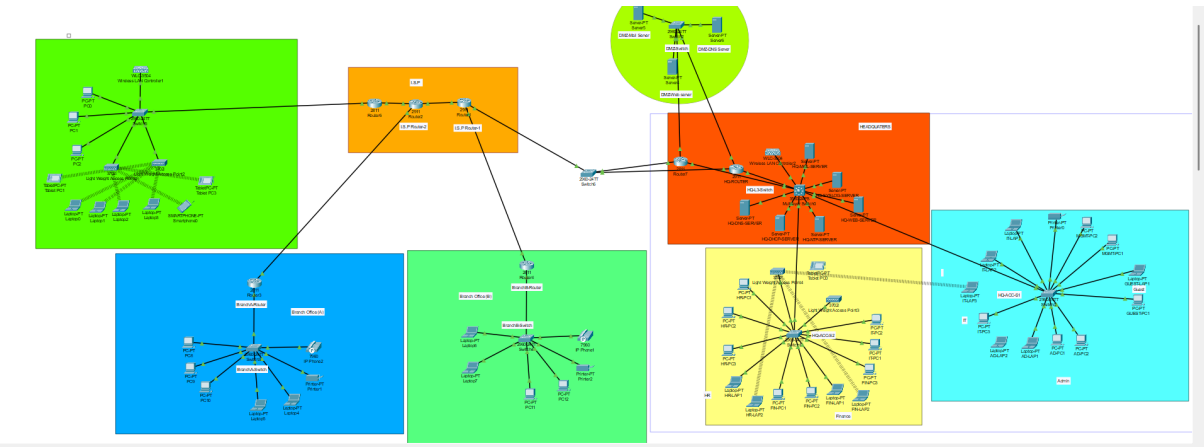
- Comprehensive security (ACLs, Port Security, SSH)

- Centralized monitoring (NTP, Syslog)

TECHNOLOGIES USED

OSPF | HSRP | VLANs | NAT | ACLs | Port Security | SSH | DHCP |

Wireless LAN Controller | Centralized Logging



Section 3: NETWORK ARCHITECTURE

SITES

- Headquarters: 8 VLANs, redundant routers, L3 switch, DMZ, wireless

- Branch A: Single router, switch, 8 PCs

- Branch B: Single router, switch, 7 PCs

- WAN Backbone: 3 ISP routers with OSPF

DEVICE COUNT

Routers: 6 | Switches: 6 | Servers: 9 | Wireless: WLC + 2 APs

DESIGN PRINCIPLES

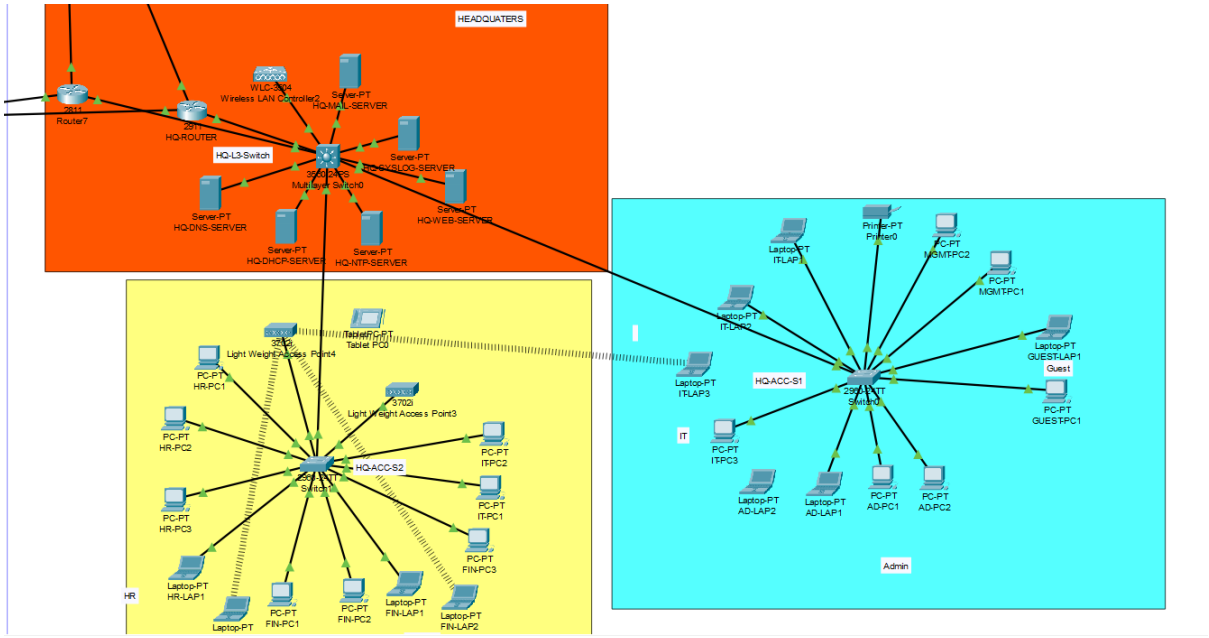
✓ Hierarchical (Core-Distribution-Access)

✓ Redundancy (HSRP at HQ)

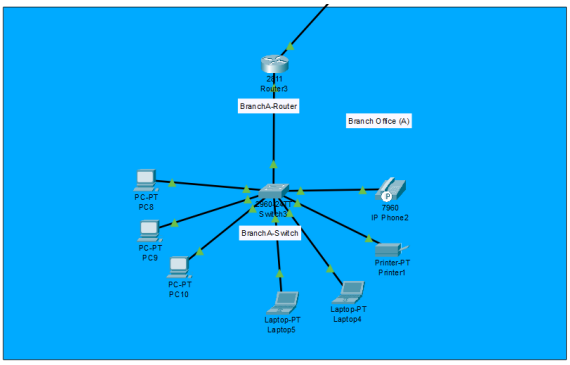
✓ Security (Multi-layer defense)

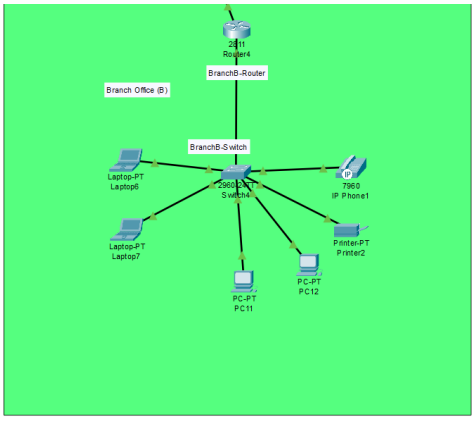
✓ Scalability (Easy to expand)

Headquarter

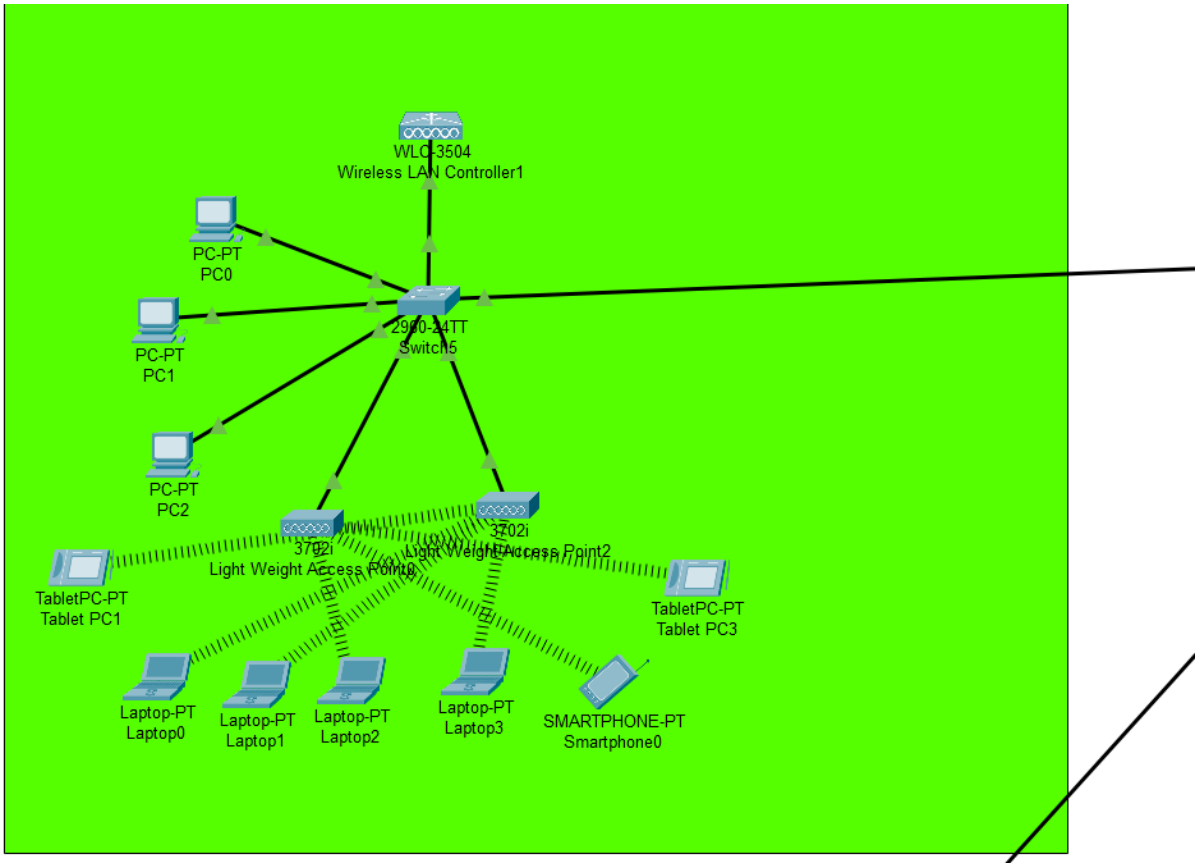


Branches

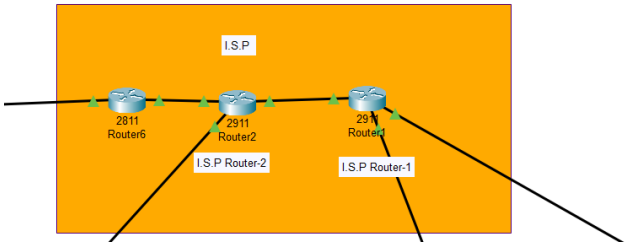


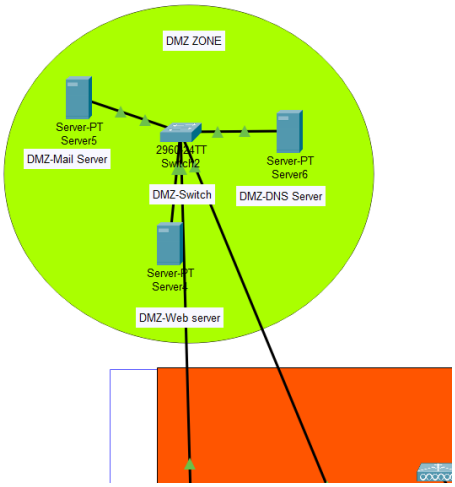


External Network



DMZ and ISP





Section 4: IP ADDRESSING SCHEME

HQ VLANs (192.168.x.0/24)

VLAN 10 - HR | 192.168.10.0/24

VLAN 20 - Finance | 192.168.20.0/24

VLAN 30 - IT | 192.168.30.0/24

VLAN 40 - Admin | 192.168.40.0/24

VLAN 50 - Guest | 192.168.50.0/24

VLAN 90 - Wireless | 192.168.90.0/24

VLAN 99 - Management | 192.168.99.0/24

VLAN 100 - Voice (Ready)| 192.168.100.0/24

BRANCHES

Branch A: 192.168.100.0/24

Branch B: 192.168.101.0/24

DMZ & EXTERNAL

DMZ Internal: 192.168.200.0/24 → Public IPs: 200.0.0.10-12 (NAT)

Guest WiFi: 200.100.0.0/24

WAN LINKS

All /30 or /29 subnets (100.0.0.0/8)

SPECIAL ADDRESSES

HSRP Virtual IP: 192.168.99.254

Servers: 192.168.99.10-15 (DHCP, DNS, Web, Email, NTP, Syslog)

Section 5: OSPF DYNAMIC ROUTING

WHY OSPF?

✓ Automatic route learning (no manual static routes)

✓ Fast convergence (3-5 seconds on failures)

✓ Load balancing across equal-cost paths

✓ Scalable and standards-based

CONFIGURATION

Area: 0 (Backbone)

All routers and L3 switch participate

Unique Router IDs: 1.1.1.1, 2.2.2.2, 3.3.3.3, etc.

HOW IT WORKS

1. Routers exchange topology information

2. SPF algorithm calculates best paths

3. Automatic failover if link fails

4. Branches can reach HQ and each other seamlessly

EXAMPLE

Branch A → Branch B traffic automatically routed via OSPF

If ISP Router 1 fails, OSPF recalculates alternate path

```
ISP-Router-1>en
ISP-Router-1#show ip route ospf
     100.0.0.0/8 is variably subnetted, 7 subnets, 3 masks
O       100.0.3.0 [110/2] via 100.0.1.2, 08:24:29, GigabitEthernet0/1
O    192.168.10.0 [110/3] via 100.0.0.3, 08:24:19, GigabitEthernet0/0
                  [110/3] via 100.0.0.2, 08:24:19, GigabitEthernet0/0
O    192.168.20.0 [110/3] via 100.0.0.3, 08:24:19, GigabitEthernet0/0
                  [110/3] via 100.0.0.2, 08:24:19, GigabitEthernet0/0
O    192.168.30.0 [110/3] via 100.0.0.3, 08:24:19, GigabitEthernet0/0
                  [110/3] via 100.0.0.2, 08:24:19, GigabitEthernet0/0
O    192.168.40.0 [110/3] via 100.0.0.3, 08:24:19, GigabitEthernet0/0
                  [110/3] via 100.0.0.2, 08:24:19, GigabitEthernet0/0
O    192.168.50.0 [110/3] via 100.0.0.3, 08:24:19, GigabitEthernet0/0
                  [110/3] via 100.0.0.2, 08:24:19, GigabitEthernet0/0
O    192.168.60.0 [110/3] via 100.0.0.3, 08:24:19, GigabitEthernet0/0
                  [110/3] via 100.0.0.2, 08:24:19, GigabitEthernet0/0
O    192.168.90.0 [110/3] via 100.0.0.3, 08:24:19, GigabitEthernet0/0
                  [110/3] via 100.0.0.2, 08:24:19, GigabitEthernet0/0
O    192.168.99.0 [110/2] via 100.0.0.3, 08:24:29, GigabitEthernet0/0
                  [110/2] via 100.0.0.2, 08:24:29, GigabitEthernet0/0
O    192.168.100.0 [110/3] via 100.0.1.2, 08:24:29, GigabitEthernet0/1
O    192.168.101.0 [110/2] via 100.0.2.2, 08:24:29, GigabitEthernet0/2
O    192.168.200.0 [110/2] via 100.0.0.2, 08:24:29, GigabitEthernet0/0
     200.0.0.0/24 is variably subnetted, 4 subnets, 2 masks
O       200.0.0.0 [110/2] via 100.0.1.2, 08:24:29, GigabitEthernet0/1
O    200.100.0.0 [110/3] via 100.0.1.2, 08:24:29, GigabitEthernet0/1

ISP-Router-1#
```

Section 6: HIGH AVAILABILITY (HSRP)

THE PROBLEM

Single router = single point of failure

If HQ router fails → entire HQ loses connectivity

THE SOLUTION

Two HQ routers sharing one virtual IP (192.168.99.254)

Router 1: Active (Priority 110)

Router 2: Standby (Priority 100)

HOW IT WORKS

Normal: Router 1 forwards all traffic

Failure: Router 2 detects failure in 10 seconds

Takeover: Router 2 becomes active automatically

Recovery: Router 1 reclaims active role when back online

TESTING RESULTS

✓ Failover successful

✓ Downtime: 10 seconds

✓ Automatic recovery working

✓ Both routers have identical configs (OSPF, NAT, ACLs)

HQ Router-1 Active

```
HQ-Router>en
HQ-Router#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Gig0/1      1    110 P Active   local           192.168.99.3    192.168.99.254
Gig0/2      2    110 P Active   local           192.168.200.2   192.168.200.254
HQ-Router#
```

HQ Router -2 Standby

```
*Mar 01, 00:00:51.000: %HSRP-6-STATECHANGE: FastEthernet0/1 Grp 1 state Speak -> Standby

HQ-Router-2>en
HQ-Router-2#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Fa0/1       1    100   Standby  192.168.99.2    local           192.168.99.254
Eth1/0      2    100   Standby  192.168.200.1   local           192.168.200.254
HQ-Router-2#
```

Section 7: VLANs & INTER-VLAN ROUTING PURPOSE

Logical segmentation by department

Reduces broadcast domains

Security through isolation

Organized network management

8 VLANS AT HQ

Departments (10, 20, 30, 40), Guest (50), Wireless (90),

Management (99), Voice-Ready (100)

INTER-VLAN ROUTING

L3 Switch acts as gateway for all VLANs

Hardware-based routing (wire speed)

EXAMPLE FLOW

HR PC (VLAN 10) → Email Server (VLAN 99)

PC sends to gateway → L3 Switch routes between VLANs → Server receives

BENEFITS

✓ Broadcasts limited to each VLAN

✓ Departments isolated

✓ Easy to add VLANs

✓ No router-on-a-stick needed

```
HQ-L3-Switch>en
HQ-L3-Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   HR                               active
20   Finance                          active
30   IT                               active
40   Admin                            active
50   Guest                            active
60   Wireless-Internal                active
90   Wireless-WLC-MGT-New             active
99   MGMT                             active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/9, Fa0/11, Fa0/12
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
HQ-L3-Switch#
```

Section 8: SECURITY - ACLs & NAT

MULTI-LAYER DEFENSE

1. Perimeter: ACLs block external → internal

2. DMZ: Isolated public services

3. Access: Port security on 34 ports

4. Management: SSH encryption

5. Physical: 144+ unused ports disabled

ACCESS CONTROL LISTS

Location: Internet Router

Rule: Block external users (200.100.0.0/24) from accessing

ALL internal networks (192.168.x.x)

Allow: External → DMZ only (200.0.0.10-12)

WHAT'S PROTECTED

HR, Finance, IT, Admin, Guest, Wireless, Management VLANs

Branch A & Branch B networks

NAT (NETWORK ADDRESS TRANSLATION)

192.168.200.10 → 200.0.0.10 (Web Server)

192.168.200.11 → 200.0.0.11 (Email Server)

192.168.200.12 → 200.0.0.12 (DNS Server)

Hides internal IPs, allows public access to DMZ

ACL

```
Internet-Router>en
Internet-Router#show access list
Internet-Router#show access-lists
Extended IP access list BLOCK-INTERNAL
    10 permit ip 200.100.0.0 0.0.0.255 200.0.0.0 0.0.0.255
    20 permit ip 200.100.0.0 0.0.0.255 host 200.0.0.10
    30 permit ip 200.100.0.0 0.0.0.255 host 200.0.0.11
    40 permit ip 200.100.0.0 0.0.0.255 host 200.0.0.12
    50 deny ip 200.100.0.0 0.0.0.255 any (137 match(es))

Internet-Router#
```

```
HQ-Router#show access-lists
Standard IP access list 1
    10 permit 192.168.200.0 0.0.0.255
Extended IP access list INTERNET-TO-INTERNAL
    10 deny ip 200.100.0.0 0.0.0.255 192.168.10.0 0.0.0.255
    20 deny ip 200.100.0.0 0.0.0.255 192.168.20.0 0.0.0.255
    30 deny ip 200.100.0.0 0.0.0.255 192.168.30.0 0.0.0.255
    40 deny ip 200.100.0.0 0.0.0.255 192.168.40.0 0.0.0.255
    50 deny ip 200.100.0.0 0.0.0.255 192.168.50.0 0.0.0.255
    60 deny ip 200.100.0.0 0.0.0.255 192.168.99.0 0.0.0.255
    70 deny ip 200.100.0.0 0.0.0.255 192.168.100.0 0.0.0.255
    80 deny ip 200.100.0.0 0.0.0.255 192.168.101.0 0.0.0.255
    90 permit ip any any (1667 match(es))

HQ-Router#
```

NAT

```
HQ-Router#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
---  200.0.0.10        192.168.200.10     ---                ---
---  200.0.0.11        192.168.200.11     ---                ---
---  200.0.0.12        192.168.200.12     ---                ---

HQ-Router#
```

Section 9: SECURITY - PORT SECURITY & SSH

PORT SECURITY

Purpose: Prevent unauthorized devices from connecting

Configuration:

- Maximum: 2 MAC addresses per port
- Violation Mode: Restrict (drop + log)
- Learning: Sticky (automatic)
- Ports Secured: 34 across all access switches

How It Works:

Legitimate device → MAC learned → Saved → Allowed

Rogue device → Violation → Packets dropped → Logged to Syslog

SSH SECURE ACCESS

All 13 devices configured with:

- SSH version 2 (encrypted)
- RSA 1024-bit keys
- Username/password authentication
- Telnet disabled

Access: ssh -l admin [device-ip]

Credentials: admin / Admin@2024

UNUSED PORTS DISABLED

144+ ports administratively shutdown

Prevents unauthorized physical access

```
HQ-Access-SW1#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
               (Count)       (Count)         (Count)
--------------------------------------------------------------------
        Fa0/2        2          1                 0         Restrict
        Fa0/3        2          1                 0         Restrict
        Fa0/4        2          1                 0         Restrict
        Fa0/5        2          1                 0         Restrict
        Fa0/6        2          1                 0         Restrict
        Fa0/7        2          1                 0         Restrict
        Fa0/8        2          1                 0         Restrict
        Fa0/9        2          1                 0         Restrict
       Fa0/10        2          1                 0         Restrict
       Fa0/11        2          1                 0         Restrict
---------------------------------------------------------------------
HQ-Access-SW1#
```

```
HQ-Access-SW2>en
HQ-Access-SW2#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
               (Count)       (Count)         (Count)
--------------------------------------------------------------------
        Fa0/2        2          1                 0         Restrict
        Fa0/3        2          1                 0         Restrict
        Fa0/4        2          1                 0         Restrict
        Fa0/5        2          1                 0         Restrict
        Fa0/6        2          1                 0         Restrict
        Fa0/7        2          1                 0         Restrict
        Fa0/8        2          1                 0         Restrict
        Fa0/9        2          1                 0         Restrict
       Fa0/10        2          1                 0         Restrict
       Fa0/11        2          1                 0         Restrict
       Fa0/12        2          1                 0         Restrict
---------------------------------------------------------------------
HQ-Access-SW2#
```

SSH

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l admin 192.168.99.2

Password:


*********************************************
   UNAUTHORIZED ACCESS PROHIBITED
   ABTechSolutions Enterprise Network
   All activities are logged and monitored
*********************************************


HQ-Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
HQ-Router(config)#
```

Section 10: WIRELESS NETWORKS

TWO SEPARATE NETWORKS

INTERNAL WIRELESS (Employees)

SSID: ABTech-Internal

Password: Internal@2024

VLAN: 90 (192.168.90.0/24)

Access: Full internal network access

Infrastructure: WLC 3504 + 2 APs

GUEST WIRELESS (Visitors)

SSID: ABTech-Guest

Password: Guest@2024

Network: 200.100.0.0/24

Access: DMZ and Internet ONLY

Blocked: All internal VLANs

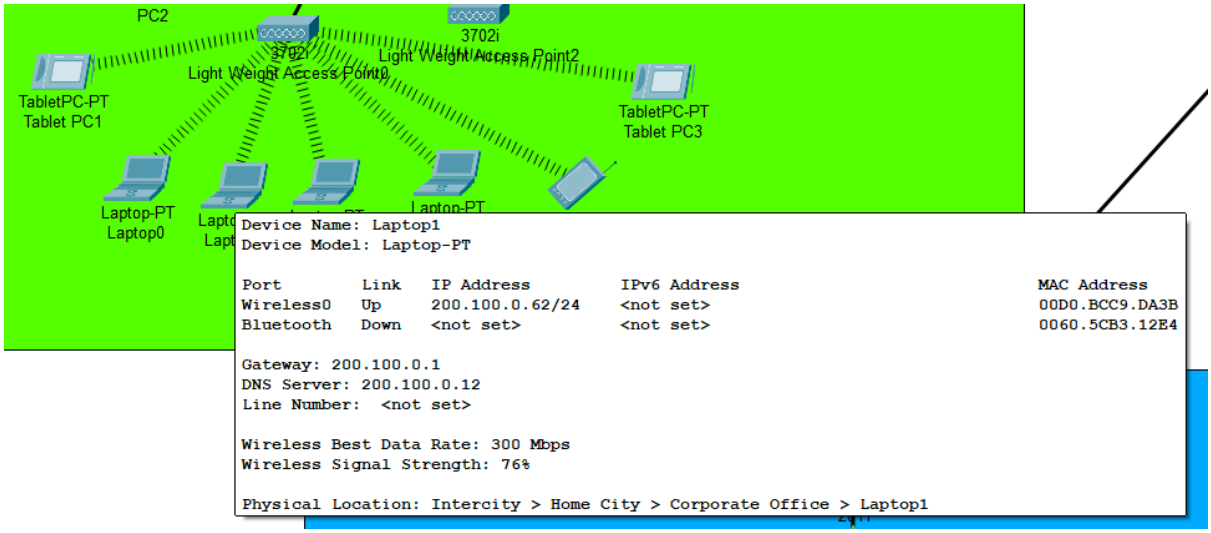
SECURITY ISOLATION

✓ Guest can access: Internet, DMZ services

✗ Guest CANNOT access: HR, Finance, IT, Admin, Branches

Enforcement: ACLs block external → internal traffic

Laptop Connected through wireless



Section 11: NETWORK MONITORING (NTP&SYSLOG)

NTP (TIME SYNCHRONIZATION)

Server: 192.168.99.14

Clients: 14 devices (all routers and switches)

Timezone: PKT (UTC+5)

Why Important:

- Log correlation across devices
- Security audit trails
- Troubleshooting efficiency
- Compliance requirements

Result: All devices synchronized within milliseconds

SYSLOG (CENTRALIZED LOGGING)

Server: 192.168.99.15

Clients: Same 14 devices

What Gets Logged:

- Configuration changes

- Interface up/down

- Port security violations

- SSH logins

- OSPF neighbor changes

Log Format:

*Dec 24 14:35:22.456: %SYS-5-CONFIG_I: Configured from console

Benefits:

✓ Centralized troubleshooting

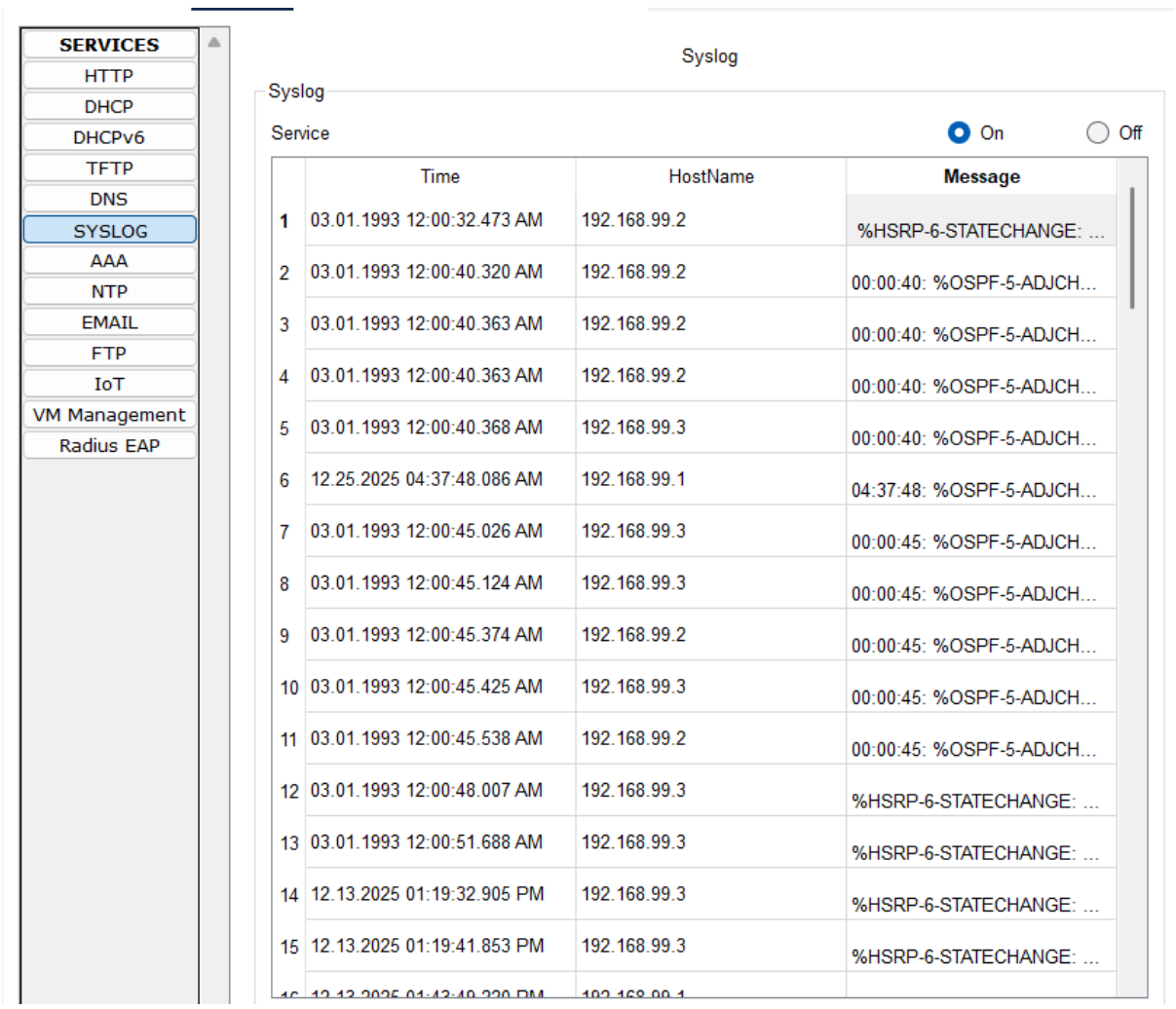
✓ Security incident investigation

✓ Real-time monitoring

NTP

```
BranchB-Router>en
BranchB-Router#show ntp status
Clock is synchronized, stratum 2, reference is 192.168.99.14
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is ECB9C8FD.00000034 (13:49:49.052 UTC Sat Dec 13 2025)
clock offset is 2.00 msec, root delay is 0.00  msec
root dispersion is 42.91 msec, peer dispersion is 0.12 msec.
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/s system poll
interval is 4, last update was 2 sec ago.
BranchB-Router#
```

SYSLOG



Section 12: DMZ & SERVER INFRASTRUCTURE

INTERNAL SERVERS (VLAN 99)

192.168.99.10 - DHCP (11 pools for all sites)

192.168.99.11 - DNS (Internal resolution)

192.168.99.12 - Web (Intranet)

192.168.99.13 - Email (Internal mail)

192.168.99.14 - NTP (Time sync)

192.168.99.15 - Syslog (Logging)

DMZ SERVERS (Public-Facing)

192.168.200.10 → 200.0.0.10 (Public Web)

192.168.200.11 → 200.0.0.11 (Public Email)

192.168.200.12 → 200.0.0.12 (Public DNS)

DMZ SECURITY

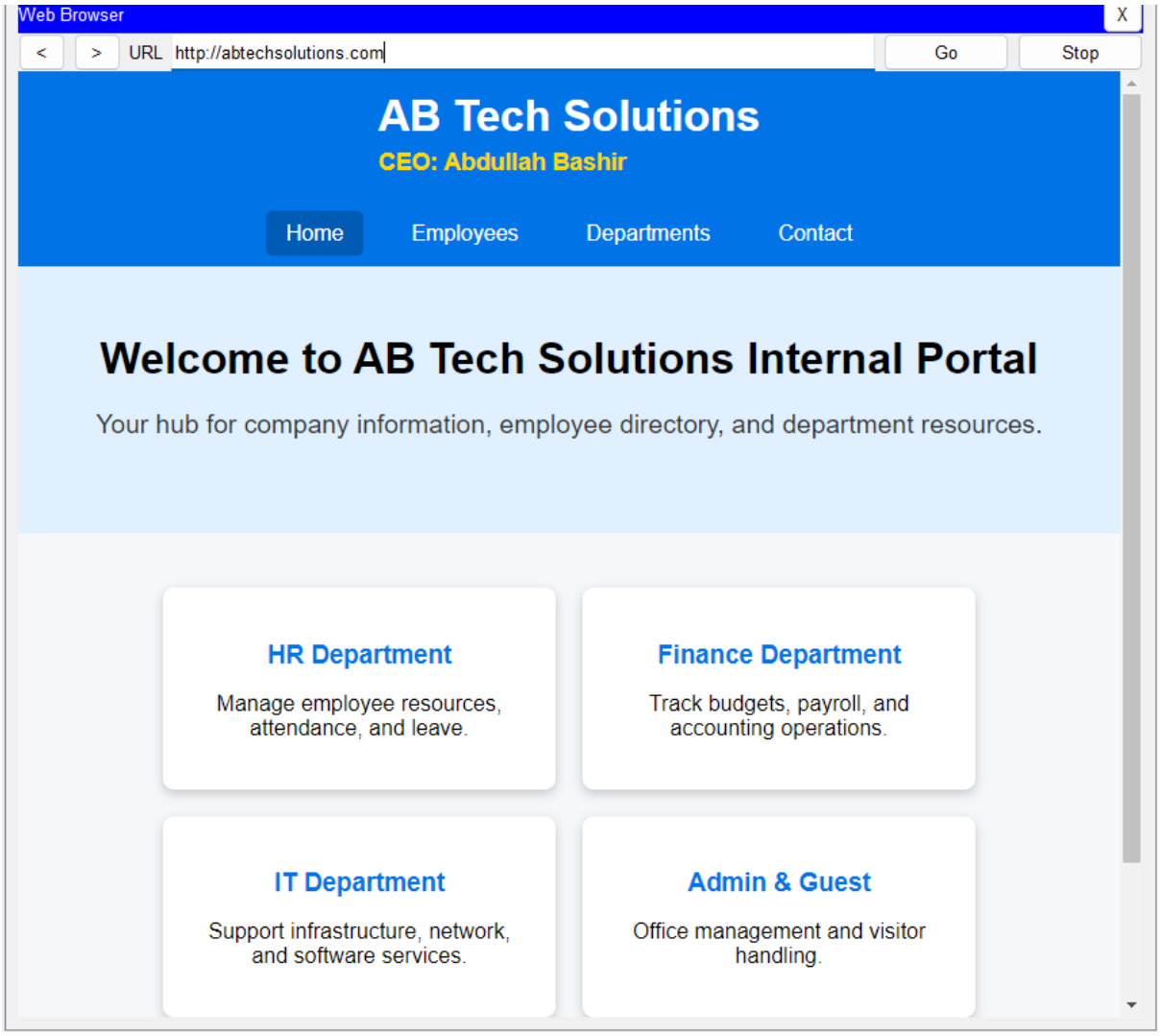
- Separate network segment
- Static NAT for public access
- ACLs control access
- Isolated from internal network
- If compromised, attacker cannot reach internal resources

TRAFFIC FLOW

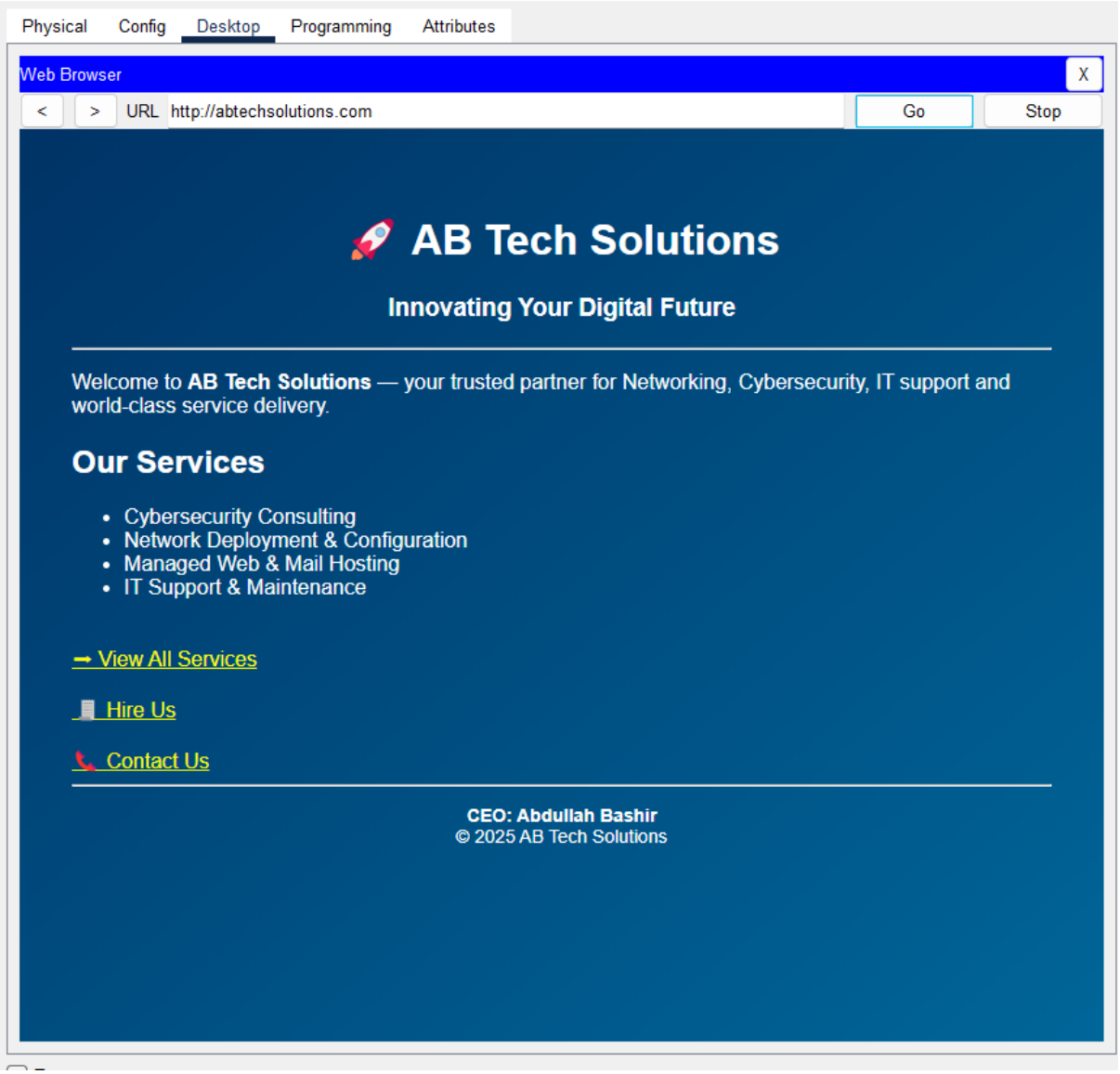
External User → Public IP → HQ Router → NAT → DMZ Server

Internal User → Can access both internal servers and DMZ

Internal Network(Employees)Website



External Network (Users) Website



PAGE 13: TESTING - CONNECTIVITY

CONNECTIVITY TESTS

✓ Test 1: Inter-VLAN Communication

HR PC → Finance PC | SUCCESS

✓ Test 2: Branch to HQ

Branch A → HQ Email Server | SUCCESS (OSPF routing working)

✓ Test 3: Branch to Branch

Branch A → Branch B | SUCCESS (Multi-hop OSPF)

✓ Test 4: External to DMZ

Guest WiFi → DMZ Web (200.0.0.10) | SUCCESS (NAT working)

SECURITY TESTS

✓ Test 5: External to Internal (Should FAIL)

Guest WiFi → HR PC (192.168.10.5) | BLOCKED ✓ (ACL working)

✓ Test 6: External to Management (Should FAIL)

Guest → L3 Switch (192.168.99.1) | BLOCKED ✓ (Protected)

HSRP FAILOVER TEST

Action: Powered off HQ Router 1

Result:

- Router 2 became Active after 10 seconds
- Traffic resumed automatically
- 3-4 ping timeouts during transition
- ✓ HSRP WORKING

PORT SECURITY TEST

Action: Changed PC MAC address (simulated rogue device)

Result:

- Violation detected
- Packets dropped
- Logged to Syslog

- ✓ PORT SECURITY WORKING

OVERALL: 15/15 Tests PASSED

Test Inter Vlan Routing

```
C:\>ping 192.168.20.103

Pinging 192.168.20.103 with 32 bytes of data:

Reply from 192.168.20.103: bytes=32 time=32ms TTL=127
Reply from 192.168.20.103: bytes=32 time<1ms TTL=127
Reply from 192.168.20.103: bytes=32 time<1ms TTL=127
Reply from 192.168.20.103: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.103:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 32ms, Average = 8ms
```

Test Branch to Branch (Multi-Hop OSPF)

```
C:\>ping 192.168.101.101

Pinging 192.168.101.101 with 32 bytes of data:

Reply from 192.168.101.101: bytes=32 time=24ms TTL=124
Reply from 192.168.101.101: bytes=32 time=11ms TTL=124
Reply from 192.168.101.101: bytes=32 time<1ms TTL=124
Reply from 192.168.101.101: bytes=32 time=11ms TTL=124

Ping statistics for 192.168.101.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 24ms, Average = 11ms
```

Test : External to Internal (Should FAIL)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.5

Pinging 192.168.10.5 with 32 bytes of data:

Reply from 200.100.0.1: Destination host unreachable.
Reply from 200.100.0.1: Destination host unreachable.
Reply from 200.100.0.1: Destination host unreachable.
Reply from 200.100.0.1: Destination host unreachable.

Ping statistics for 192.168.10.5:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Failed due to ACL

HSPR Test

```
HQ-Router-2#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Fa0/1       1    100   Active   local           unknown         192.168.99.254
Eth1/0      2    100   Active   local           unknown         192.168.200.254
HQ-Router-2#
```

Standby Router activated after router 1 turned off

Section 14: CHALLENGES & SOLUTIONS

CHALLENGE 1: OSPF Router ID Conflicts

Problem: Duplicate Router IDs prevented adjacencies

Solution: Manually configured unique IDs (1.1.1.1, 2.2.2.2, etc.)

Lesson: Always manually assign OSPF Router IDs

CHALLENGE 2: DHCP at Branches Not Working

Problem: Branch PCs not getting IP addresses

Solution: Added "ip helper-address 192.168.99.10" on branch routers

Lesson: DHCP relay essential for centralized DHCP

CHALLENGE 3: HSRP Failover Failed

Problem: Router 2 couldn't route when Active

Solution: Cloned entire config from Router 1 to Router 2

Lesson: Redundant routers need identical configurations

CHALLENGE 4: Switches Couldn't Ping Servers

Problem: L2 switches couldn't reach other subnets

Solution: Configured "ip default-gateway" on each switch

Lesson: Layer 2 switches need default gateway

CHALLENGE 5: Wireless Clients in Wrong VLAN

Problem: Getting VLAN 99 IPs instead of VLAN 90

Solution: Mapped WLAN to VLAN90 interface on WLC

Lesson: WLC interface mapping is critical

CHALLENGE 6: NAT Not Working

Problem: External users couldn't access DMZ

Solution: Configured NAT on both HQ routers with inside/outside designation

Lesson: Both HSRP routers need identical NAT configs

KEY TAKEAWAYS

✓ Plan IP addressing carefully

✓ Test incrementally

✓ Redundancy requires identical configs

✓ Document everything

✓ Know your tool's limitations

Section 15: PROJECT STATISTICS & CONCLUSION

FINAL STATISTICS

Infrastructure:

- Sites: 3 (HQ + 2 Branches)
- Devices: 50+ (6 routers, 6 switches, 9 servers)
- VLANs: 8
- Subnets: 15+

Security:

- Secured Ports: 34 (Port Security)
- Disabled Ports: 144+
- ACL Rules: 15+
- NAT Translations: 3

High Availability:

- HSRP Routers: 2
- Failover Time: 10 seconds
- Uptime Design: 99.9%

Monitoring:

- NTP Clients: 14
- Syslog Clients: 14
- DHCP Pools: 11

Wireless:

- SSIDs: 2
- Access Points: 2+
- WLC:

Testing:

- Tests Performed: 15
- Tests Passed: 15 (100%)

SKILLS DEMONSTRATED

✓ Enterprise network design

✓ Routing protocols (OSPF)

✓ Switching (VLANs, Trunking)

✓ High availability (HSRP)

✓ Network security (ACLs, NAT, Port Security)

✓ Wireless networking (WLC)

✓ Centralized monitoring (NTP, Syslog)

✓ Problem-solving and troubleshooting

FUTURE ENHANCEMENTS

- VoIP with IP phones and QoS
- AAA/RADIUS authentication
- Site-to-site VPN
- Additional branch offices
- IPS/IDS deployment

CONCLUSION

This project demonstrates a complete, enterprise-grade network infrastructure with proper segmentation, security, redundancy, and monitoring. The design follows industry best practices and is ready for real-world deployment.